

MUSCAT INTL, Oman

WMO: 412560

Lat: 23.6086N

Lon: 58.2614E

Elev: 8

StdP: 101.22

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.4	16.4	-0.1	3.8	32.1	2.2	4.4	30.2	8.9	24.6	7.9	23.8	1.6	200	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.1	42.6	22.3	41.1	22.7	40.0	22.9	30.6	33.9	30.1	33.6	29.6	33.2	4.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
29.9	27.0	32.8	29.2	25.9	32.5	28.9	25.4	32.4	103.4	33.6	100.9	33.8	98.6	33.0	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.6	13.0	46.0	0.9	1.4	12.3	47.1	11.8	47.9	11.3	48.7	10.6	49.7		
			8.3	31.6	1.0	0.5	7.5	32.0	7.0	32.2	6.4	32.5	5.7	32.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	20.9	22.1	25.0	29.6	34.3	34.9	33.4	31.9	31.1	29.7	25.5	22.3
	DBStd	5.30	1.56	2.10	2.43	2.83	2.79	2.62	2.44	2.41	1.83	1.85	1.83	1.68
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	1	1	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6726	338	340	464	587	753	746	726	679	634	611	466	381
	CDD18.3	3686	81	107	206	337	494	496	468	421	384	352	216	123
	CDH23.3	49467	271	540	1731	4533	8096	8158	7277	6157	5364	4678	2025	636
	CDH26.7	29033	14	87	533	2445	5629	5762	4798	3686	2994	2417	606	62
Wind	WSAvg	2.7	2.6	2.8	2.8	2.8	3.2	3.0	3.0	2.8	2.5	2.4	2.4	2.4
Precipitation	PrecAvg	86	17	17	11	8	3	0	2	1	0	0	7	11
	PrecMax	240	101	86	145	67	69	10	68	28	2	13	76	112
	PrecMin	1	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	65	23	24	24	15	11	1	10	4	0	2	15	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	31.8	35.2	40.3	44.8	44.9	43.1	42.1	39.9	38.6	33.3	29.9
		MCWB	18.6	18.3	19.2	20.5	21.5	22.4	23.3	23.3	22.5	21.1	20.8	19.1
	2%	DB	26.8	29.0	32.5	38.1	42.2	42.8	41.0	39.9	37.9	36.8	31.6	28.1
		MCWB	19.2	19.4	19.0	20.3	21.4	22.3	24.0	24.4	23.3	21.6	21.8	20.7
	5%	DB	25.8	27.2	30.9	36.2	40.9	41.1	39.3	38.0	36.1	35.1	30.5	27.1
		MCWB	19.4	19.7	19.4	20.6	21.7	22.9	24.3	24.8	24.4	22.4	22.8	20.8
	10%	DB	24.9	26.1	29.5	34.9	39.2	39.9	37.8	35.9	34.7	33.9	29.8	26.2
		MCWB	19.2	19.8	19.9	20.7	21.9	23.3	25.5	26.3	25.4	23.2	22.8	20.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	23.3	25.3	27.1	29.9	31.1	31.2	30.6	29.9	28.6	25.7	23.5
		MCDB	25.3	26.2	28.2	31.2	33.5	34.5	34.4	34.0	32.9	32.1	29.9	26.9
	2%	WB	21.2	22.5	24.2	26.1	29.0	30.5	30.7	29.9	29.2	27.7	25.0	22.5
		MCDB	24.6	25.6	27.3	30.2	33.1	34.0	33.9	33.3	32.4	31.4	29.4	26.2
	5%	WB	20.6	21.8	23.5	25.3	28.3	30.0	30.1	29.3	28.7	27.0	24.2	21.9
		MCDB	24.2	25.2	26.7	29.7	32.5	33.7	33.5	32.8	31.9	31.0	28.9	25.7
	10%	WB	19.7	21.1	22.7	24.5	27.4	29.5	29.7	28.9	28.2	26.5	23.6	21.1
		MCDB	23.7	24.4	26.3	29.4	32.4	33.4	33.2	32.3	31.3	30.7	28.3	25.1
Mean Daily Temperature Range	5% DB	MDBR	7.7	7.8	8.0	9.1	9.3	8.1	6.6	6.6	7.4	8.8	8.1	7.8
		MCDBR	8.8	9.3	10.3	11.3	11.1	10.5	9.6	9.7	10.0	10.7	9.0	8.7
		MCWBR	5.5	6.0	7.0	7.3	7.3	8.2	7.3	6.6	7.1	8.1	6.5	5.4
	5% WB	MCDBR	8.0	7.7	7.6	8.7	8.3	7.6	6.7	7.0	7.3	7.8	7.9	7.8
		MCWBR	5.4	5.3	5.9	7.0	7.4	7.0	5.4	4.9	5.4	6.4	6.0	5.1
Clear-Sky Solar Irradiance	taub	0.423	0.468	0.529	0.578	0.602	0.674	0.746	0.686	0.582	0.482	0.443	0.424	
	taud	2.229	2.073	1.900	1.817	1.768	1.658	1.571	1.664	1.833	2.044	2.182	2.218	
	Ebn at Noon	828	814	782	752	730	674	628	666	728	789	800	809	
	Edh at Noon	127	158	197	218	228	252	275	250	207	160	132	124	
All-Sky Solar Radiation	RadAvg	4.46	5.26	6.12	6.80	7.45	7.29	7.07	6.62	6.24	5.64	4.69	4.22	
	RadStd	0.22	0.24	0.32	0.36	0.17	0.25	0.28	0.18	0.12	0.15	0.14	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.23	N/A	N/A	N/A	+0.39	+0.45	N/A	N/A	+96	+98
(47)	Regional (12 neighbors)	+0.42	+0.68	N/A	N/A	+0.28	+0.30	N/A	N/A	+96	+98

Nomenclature: See separate page